Mathematical and Computational Methods

Exercise Sheet 3: Complex numbers II: exponential form, De Moivre, roots

This sheet accompanies *Unit 3 (Complex numbers II: exponential form, De Moivre, roots)*. It exercises the exponential form, De Moivre's theorem, the nth roots of a complex number, and the trigonometric identities they generate; nothing beyond Units 1–3 is needed.

How to use this sheet. Problems are graded by difficulty—$\star$ drill (routine fluency), $\star\star$ core (standard, tutorial level) and $\star\star\star$ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit3}"`) and recompile.

Notation. As in the notes, $\mathrm{i}^2 = -1$, $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$, a nonzero z has exponential form $z = r\,\mathrm{e}^{\mathrm{i}\theta}$ with $r = |z|$ and $\theta = \arg z$, and $\bar{z}$ is the complex conjugate. Give exact answers (integers, simple surds or fractions) unless a decimal is asked for.

Contents

1 Warm-up drills

Short routine questions, at least one per skill. Aim for speed and accuracy; a few ask for a one- or two-line "show" derivation.

Exercise 1.1 *($\star$ drill— exponential form)*. Write each number in exponential form $r\,\mathrm{e}^{\mathrm{i}\theta}$ with $r > 0$ and $-\pi < \theta \le \pi$:

$$\text{(a) } 1+\mathrm{i}, \quad \text{(b) } -3, \quad \text{(c) } 2\mathrm{i}, \quad \text{(d) } -1-\mathrm{i}, \quad \text{(e) } \sqrt{3}-\mathrm{i}, \quad \text{(f) } -2+2\sqrt{3}\,\mathrm{i}.$$

Exercise 1.2 *($\star$ drill— back to Cartesian)*. Write each in Cartesian form $a + \mathrm{i}b$:

$$\text{(a) } 2\,\mathrm{e}^{\mathrm{i}\pi/3}, \quad \text{(b) } 3\,\mathrm{e}^{\mathrm{i}\pi}, \quad \text{(c) } \sqrt{2}\,\mathrm{e}^{-\mathrm{i}\pi/4}, \quad \text{(d) } 4\,\mathrm{e}^{\mathrm{i}3\pi/2}, \quad \text{(e) } \mathrm{e}^{\mathrm{i}5\pi/6}.$$

Exercise 1.3 *(⋆ drill— Euler values)***.** Evaluate $e^{i\pi/2}$, $e^{i2\pi}$, $e^{-i\pi}$ and $e^{i\pi/3}$, and state Euler's identity.

Exercise 1.4 *(⋆ drill— rotation)***.** Let $z = 2\,e^{i\pi/6}$. Write down $e^{i\pi/2}z$ in exponential form, and describe geometrically the effect of multiplying a complex number by (i) i, (ii) $e^{i\pi}$, (iii) $e^{-i\pi/2}$.

Exercise 1.5 *(⋆ drill— De Moivre)***.** Use De Moivre's theorem to simplify $(\cos\theta + i\sin\theta)^4$ and $(\cos\theta + i\sin\theta)^{-2}$.

Exercise 1.6 *(⋆ drill— integer powers)***.** Compute, using the exponential form:

$$\text{(a) } (1+i)^4, \quad \text{(b) } (\sqrt{3}+i)^3, \quad \text{(c) } (1-i)^6, \quad \text{(d) } \left(2\,e^{i\pi/5}\right)^{10}.$$

Exercise 1.7 *(⋆ drill— a few roots)***.** Find all the (a) square roots of i, and (b) fourth roots of 1.

Exercise 1.8 *(⋆ drill— Argand sketch)***.** Plot the four fourth roots of unity $1, i, -1, -i$ on an Argand diagram, mark the unit circle, and name the regular polygon they form.

Exercise 1.9 *(⋆ drill— counting roots (FTA))***.** State, with the fundamental theorem of algebra in mind, how many roots in $\mathbb{C}$ (counted with multiplicity) each has: (a) a degree-5 polynomial; (b) $z^7 - 3 = 0$; (c) $(z-1)^2(z^2+1) = 0$. True or false: every real cubic has at least one real root?

Exercise 1.10 *(⋆ drill— double angle by De Moivre)***.** By expanding $(\cos\theta + i\sin\theta)^2$, derive the identities for $\cos 2\theta$ and $\sin 2\theta$.

Exercise 1.11 *(⋆ drill— hyperbolic link)***.** Using $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, evaluate $\cos(i)$ in terms of e. *(The general identity* $\cos(ix) = \cosh x$ *is built up in the tutorial.)*

Exercise 1.12 *(⋆ drill— phasors)***.** (a) Write $5\cos(\omega t + \frac{\pi}{3})$ and $2\sin\omega t$ as phasors $P = A\,e^{i\varphi}$. (b) The phasor $3\,e^{-i\pi/2}$ represents which real signal $A\cos(\omega t + \varphi)$?

2 Tutorial problems

The core set, to be worked through in the tutorial. Anchored on the worked examples of the notes. Problems marked ▶ are the core set for the 2-hour tutorial; the rest are for completion at home.

Exercise 2.1 *(▶ ⋆⋆ core— arithmetic in exponential form)***.** Let $z = \sqrt{3} + i$ and $w = 1 + i$. Write z and w in exponential form, then find zw and z/w in *both* exponential and Cartesian form.

Exercise 2.2 *(⋆⋆ core— rotation in action)***.** (a) Rotate $z = 3 + 4i$ anticlockwise by $\frac{\pi}{2}$ about the origin, and then (separately) clockwise by $\frac{\pi}{2}$. (b) Rotate $2 + 2i$ anticlockwise by $\frac{\pi}{4}$. Give exact Cartesian answers.

Exercise 2.3 *(⋆⋆ core— prove De Moivre)***.** Prove by induction that $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ for all integers $n \geq 1$. (Pure-maths.)

Exercise 2.4 *(▶ ⋆⋆ core— powers by De Moivre* (anchor)*)***.** Compute $(1+i)^8$ and $(\sqrt{3}+i)^6$ using De Moivre's theorem.

Exercise 2.5 *(⋆⋆ core— a higher power)***.** Evaluate $(-1+i)^{10}$, and confirm your answer by squaring $-1+i$ first.

Exercise 2.6 *(▶ ⋆⋆ core— roots, two shapes* (anchor)*)***.** Find (a) the cube roots of 8 and (b) the fourth roots of -16. In each case give exact Cartesian values and describe the polygon they form on the Argand diagram.

Exercise 2.7 *(▶ ⋆⋆ core— sixth roots of unity* (anchor)*)*. Find all sixth roots of unity in Cartesian form, and verify that they sum to zero. Why does this generalise to all $n \geq 2$?

Exercise 2.8 *(⋆⋆ core— solving $w^n = z$)*. Solve $z^3 = -27\mathrm{i}$, giving the three roots in exponential and Cartesian form.

Exercise 2.9 *(▶ ⋆⋆ core—* $\cos 3\theta$ *and* $\sin 3\theta$ (anchor)*)*. Using De Moivre and the binomial theorem, show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ and $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$.

Exercise 2.10 *(⋆⋆ core— power reduction)*. Use the inverse Euler formula to write $\sin^4\theta$ as a sum of cosines of multiples of θ.

Exercise 2.11 *(▶ ⋆⋆ core— trig–hyperbolic link)*. (a) Show $\cos(\mathrm{i}x) = \cosh x$ and $\sin(\mathrm{i}x) = \mathrm{i}\sinh x$. (b) Hence evaluate $\cos(\mathrm{i}\ln 2)$ exactly.

Exercise 2.12 *(▶ ⋆⋆ core— combining oscillations* (anchor; physics)*)*. (a) Combine $3\cos\omega t + 4\sin\omega t$ into a single sinusoid $R\cos(\omega t - \alpha)$. (b) Combine $2\cos\omega t + 2\cos(\omega t + \frac{\pi}{2})$ likewise.

3 Further practice (home)

A larger bank for independent study, routine to hard, with challenge problems marked ⋆⋆⋆.

Euler, forms and rotation (Skills 1–2)

Exercise 3.1 *(⋆ drill— mixed conversions)*. (a) To exponential form ($-\pi < \theta \leq \pi$): -4, $3\mathrm{i}$, $-2-2\mathrm{i}$, $-1+\sqrt{3}\,\mathrm{i}$. (b) To Cartesian form: $5\,\mathrm{e}^{\mathrm{i}\pi/2}$, $2\,\mathrm{e}^{-\mathrm{i}2\pi/3}$, $\sqrt{2}\,\mathrm{e}^{\mathrm{i}3\pi/4}$.

Exercise 3.2 *(⋆ drill— index laws)*. Simplify to a single $r\,\mathrm{e}^{\mathrm{i}\theta}$ (then to Cartesian if clean): (a) $\mathrm{e}^{\mathrm{i}\pi/3}\cdot\mathrm{e}^{\mathrm{i}\pi/6}$, (b) $\dfrac{8\,\mathrm{e}^{\mathrm{i}5\pi/6}}{2\,\mathrm{e}^{\mathrm{i}\pi/3}}$, (c) $\left(\sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/8}\right)^4$.

Exercise 3.3 *(⋆⋆ core— a rotated square)*. A square is centred at the origin with one vertex at $z_0 = 2 + \mathrm{i}$. Find its other three vertices by repeatedly multiplying by i, and state the side length.

Exercise 3.4 *(⋆⋆ core— rotation preserves geometry)*. Let $\alpha \in \mathbb{R}$. Show that for all $z, w \in \mathbb{C}$, $|\mathrm{e}^{\mathrm{i}\alpha}z| = |z|$ and $|\mathrm{e}^{\mathrm{i}\alpha}z - \mathrm{e}^{\mathrm{i}\alpha}w| = |z - w|$. What does the second identity say geometrically?

De Moivre and powers (Skills 3–4)

Exercise 3.5 *(⋆⋆ core— more powers)*. Evaluate (a) $(1-\mathrm{i}\sqrt{3})^9$, (b) $(2+2\mathrm{i})^5$, (c) $\left(\dfrac{1+\mathrm{i}}{\sqrt{3}+\mathrm{i}}\right)^{12}$.

Exercise 3.6 *(⋆⋆ core— De Moivre for negative n)*. Assuming De Moivre for positive integers, prove it for negative integers $n = -m$ ($m \in \mathbb{N}$). [Hint: $(\cos\theta + \mathrm{i}\sin\theta)^{-1} = \cos\theta - \mathrm{i}\sin\theta$.]

Roots and the fundamental theorem (Skills 5–6)

Exercise 3.7 *(⋆ drill— assorted roots)*. Find all: (a) square roots of $2\mathrm{i}$; (b) cube roots of -1; (c) fourth roots of -1; (d) fifth roots of 32 (leave in exponential form).

Exercise 3.8 *(⋆⋆ core— square root the algebraic way)*. The argument of $5 - 12\mathrm{i}$ is not a standard angle, so the exponential method is awkward. Instead find both square roots of $5 - 12\mathrm{i}$ in Cartesian form by equating real and imaginary parts of $(a + ib)^2$, and verify by squaring.

Exercise 3.9 *(⋆⋆ core— a non-real right-hand side)*. Solve $z^3 = 1 + \mathrm{i}$, leaving the roots in exponential form.

Exercise 3.10 *(⋆⋆ core— a sixth-degree equation)*. Solve $z^6 = 64$, giving all six roots in Cartesian form, and describe their configuration.

Exercise 3.11 *(⋆⋆ core— data-science flavour: roots of unity)*. In an N-point discrete Fourier transform the sampling frequencies are the Nth roots of unity. Take $N = 4$ and $\omega = \mathrm{e}^{\mathrm{i}2\pi/4}$. (a) List $\omega^0, \omega^1, \omega^2, \omega^3$. (b) Show that $\sum_{k=0}^{3} \omega^k = 0$ and $\sum_{k=0}^{3} \omega^{2k} = 0$, whereas $\sum_{k=0}^{3} \omega^{0\cdot k} = 4$. [These "orthogonality" sums are what make the DFT invertible.]

Exercise 3.12 *(⋆⋆ core— data-science flavour: twiddle factors)*. The fast Fourier transform is built from the "twiddle factors" $W_N^k = \mathrm{e}^{-\mathrm{i}2\pi k/N}$. Take $N = 8$. (a) Write W_8^1 in Cartesian form. (b) Show that $W_8^{k+4} = -W_8^k$ for every integer k (the symmetry that halves the FFT's work). (c) Give the modulus and argument of W_8^3.

Exercise 3.13 *(⋆⋆ core— FTA and real factors)*. (a) Verify that $z = 2\mathrm{i}$ is a root of $z^2 + 4$, and write down the other. (b) Factor $z^4 + 4$ into two real quadratics. [Hint: find the four roots first.]

Exercise 3.14 *(⋆ ⋆ ⋆ challenge— a root-of-unity product)*. Let $\omega = \mathrm{e}^{\mathrm{i}2\pi/5}$. Show that $(1-\omega)(1-\omega^2)(1-\omega^3)(1-\omega^4) = 5$. Generalise to n.

Exercise 3.15 *(⋆ ⋆ ⋆ challenge— a conjugate equation)*. Find all $z \in \mathbb{C}$ with $z^2 = \bar{z}$.

Trigonometric identities (Skill 7)

Exercise 3.16 *(⋆⋆ core— $\cos 4\theta$)*. Use De Moivre to show $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$.

Exercise 3.17 *(⋆⋆ core— $\cos^5\theta$)*. Use the inverse Euler formula to express $\cos^5\theta$ as a sum of cosines of multiples of θ.

Exercise 3.18 *(⋆ ⋆ ⋆ challenge— $\tan 3\theta$)*. Show that $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$.

Exercise 3.19 *(⋆ ⋆ ⋆ challenge— a finite cosine sum)*. Using the geometric series for $\sum_{k=0}^{n-1} \mathrm{e}^{ik\theta}$ (with $\theta \neq 0 \bmod 2\pi$), show that

$$\sum_{k=0}^{n-1} \cos k\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\tfrac{(n-1)\theta}{2}\right).$$

Trigonometric–hyperbolic link (Skill 8)

Exercise 3.20 *(⋆⋆ core— the converse pair)*. Show that $\cosh(\mathrm{i}x) = \cos x$ and $\sinh(\mathrm{i}x) = \mathrm{i}\sin x$, and evaluate $\sin(2\mathrm{i})$ and $\cos(\pi\mathrm{i})$ in terms of e.

Exercise 3.21 *(⋆⋆ core— identity transfer)*. Starting from $\cos 2\theta = 1 - 2\sin^2\theta$, set $\theta = \mathrm{i}x$ and use the link of the previous exercise to derive the hyperbolic identity $\cosh 2x = 1 + 2\sinh^2 x$.

Phasors (Skill 9)

Exercise 3.22 *(⋆⋆ core— a 5–12–13 combination)*. Combine $5\cos\omega t + 12\sin\omega t$ into $R\cos(\omega t - \alpha)$, giving R exactly and α to three decimals.

Exercise 3.23 *(⋆⋆ core— physics word problem: two AC sources)*. Two alternating voltages of equal amplitude but different phase, $V_1 = 10\cos\omega t$ and $V_2 = 10\cos(\omega t + \frac{2\pi}{3})$ (volts), are added. Find the amplitude and phase of $V_1 + V_2$.

Exercise 3.24 *($\star\star\star$ challenge— balanced three-phase)***.** Show that three equal phasors $120°$ apart sum to zero: $1 + \mathrm{e}^{\mathrm{i}2\pi/3} + \mathrm{e}^{\mathrm{i}4\pi/3} = 0$. Interpret the result physically.

Concept checks (true/false and spot-the-error)

Exercise 3.25 *($\star$ drill— true or false)***.** Decide and justify briefly: (a) $\left|\mathrm{e}^{\mathrm{i}\theta}\right| = 1$ for every real θ. (b) $\mathrm{e}^{\mathrm{i}\theta} = \mathrm{e}^{\mathrm{i}\varphi}$ implies $\theta = \varphi$. (c) Every complex number has exactly three cube roots. (d) The nth roots of unity always include 1.

Exercise 3.26 *($\star\star$ core— spot the error)***.** A student writes: "Since $(\cos\theta + \mathrm{i}\sin\theta)^{1/2} = \cos\frac{\theta}{2} + \mathrm{i}\sin\frac{\theta}{2}$ by De Moivre, the square root of $\mathrm{e}^{\mathrm{i}\theta}$ is $\mathrm{e}^{\mathrm{i}\theta/2}$." What is wrong, and what is the correct statement?